

Rational Expressions Practice

MCR3U

1. Simplify and state the restrictions on the variable.

a) $\frac{2x^3}{3x^2 - 4x} \times \frac{6x - 8}{2x - 2}$

b) $\frac{2x^2 - 3x}{x^2} \times \frac{x^2 - 3x - 4}{2x^2 - x - 3}$

c) $\frac{2x^2 - 13x + 20}{x} \times \frac{x^2 + 4x - 5}{x^2 - 5x + 4}$

d) $\frac{x^2 - 1}{x^2 - 4} \times \frac{x^2 - 3x + 4}{x^2 + 5x + 4}$

2. Simplify and state the restrictions on the variable.

a) $\frac{x^2 - 4}{x + 3} \div \frac{x^2 - x - 6}{x^2 + x - 6}$

b) $\frac{x^2 + 2x - 8}{x^2 + x - 6} \div \frac{x^2 + 3x - 4}{x^2 - 9}$

c) $\frac{x^2 + x}{3x^2 - 14x + 8} \div \frac{x^2 + 3x}{3x^2 - 10x - 8}$

d) $\frac{x^3 + 4x^2 + 3x}{2x^2 - 5x - 12} \div \frac{x^2 + 3x}{2x^2 + 9x + 9}$

3. Simplify and state the restrictions on the variable.

a) $\frac{4}{x^2 - x - 6} + \frac{2}{x^2 - 2x - 3}$

b) $\frac{2x}{x^2 + x - 12} - \frac{x}{x^2 - 9}$

c) $\frac{3x - 1}{2x^2 - 7x - 4} + \frac{2x - 3}{2x^2 + 3x + 1}$

d) $\frac{2x - 1}{x^2 + 3x + 2} - \frac{2x + 2}{x^2 + 5x + 6}$

4. Remove a common factor of -1 from one of the denominators and then simplify the expressions. Be sure to state any restrictions for the variable.

a) $\frac{4}{2x - 1} + \frac{2}{1 - 2x}$

b) $\frac{3x}{x - 7} - \frac{x}{7 - x}$

c) $\frac{6x - 1}{3x - 4} + \frac{4x - 1}{4 - 3x}$

d) $\frac{5x + 2}{5x - 2} - \frac{3x}{2 - 5x}$